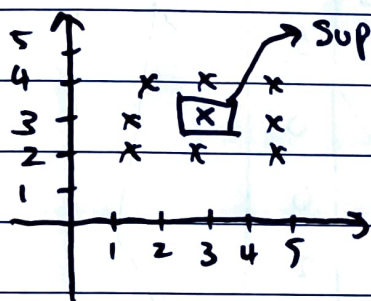


Handwritten representation exercise for DIF.

Input datapoints:

Chosen so that we have this graph. Why? To see if the randomly generated n.n can successfully transform the dataset & isolate the point in the middle of the circle.



Supposed anomaly (To be isolated).

(But in 3D & in range $[-5, 5]$).

Datapoints:

1. $[0, -5, 5]$.
2. $[-2, -2, 3]$.
3. $[-5, 0, 4]$.
4. $[-3, 3, -2]$.
5. $[0, 5, -3]$.
6. $[2, 3, -1]$.
7. $[5, 0, 0]$.
8. $[3, -2, -5]$.
9. $[0, 0, 0] \rightarrow$ To be isolated

$$D = \begin{bmatrix} 0 & -5 & 5 \\ -2 & -2 & 3 \\ -5 & 0 & 4 \\ -3 & 3 & -2 \\ 0 & 5 & -3 \\ 2 & 3 & -1 \\ 5 & 0 & 0 \\ 3 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Representation 1:

- Only 1 hidden layer

- $h = \text{ReLU}(w_1 x + b_1) \rightarrow \text{Hidden layer 1}$

- $o = (w_2 h + b_2) \rightarrow \text{output layer.}$

randomly selected values for w, q, p, b are in range $[-2, 2]$.

$$W_{0,1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix} \quad P_0 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad q_0 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$$W_1 = W_{0,1} \circ P_0 q_0^T = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 2 & 4 \\ -4 & 0 & -4 \end{bmatrix}.$$

$$b_1 = [1, -2, 0].$$

$$P_{\text{output}} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad q_{\text{output}} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad W_{0,2} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & -2 \end{bmatrix}$$

$$W_2 = W_{0,2} \circ P_{\text{out}} q_{\text{out}}^T = \begin{bmatrix} 0 & 1 & -1 \\ 8 & 0 & 2 \\ 2 & -1 & 2 \end{bmatrix}.$$

$$b_2 = [2, 1, -1].$$

Passing the dataset through representation 1:

D = dataset (input)

W_1 = ~~the~~ hidden layer 1 weights.

b_1 = hidden layer 1 bias.

W_2 = output layer weights.

b_2 = output layer weights.

O = output dataset

h = data after passing hidden layer

$$h = \text{ReLU}(W_1 D + b_1)$$

$$O = (W_2 h + b_2)$$

$$W_1 D + b_1 = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 2 & 4 \\ -4 & 0 & -4 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$
$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & -1 & 6 & 10 & 11 & 5 & -4 & -6 & 1 \\ 8 & 10 & 24 & 2 & -4 & -4 & -12 & -32 & -2 \\ -20 & -4 & 4 & 20 & 12 & -4 & -20 & 8 & 0 \end{bmatrix}$$

$$h = \text{ReLU}(W_1 D + b_1)$$

$$= \begin{bmatrix} 0 & 0 & 6 & 10 & 11 & 5 & 0 & 0 & 1 \\ 8 & 10 & 24 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 20 & 12 & 0 & 0 & 8 & 0 \end{bmatrix}$$

$$O = \begin{bmatrix} 0 & 1 & -1 \\ 8 & 0 & 2 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 6 & 10 & 11 & 5 & 0 & 0 & 1 \\ 8 & 10 & 24 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 20 & 12 & 0 & 0 & 8 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$O = \begin{bmatrix} 10 & 12 & 22 & -16 & -10 & 2 & 2 & -6 & 2 \\ 1 & 1 & 57 & 121 & 113 & 41 & 1 & 17 & 9 \\ -9 & 11 & -5 & 57 & 45 & 9 & -1 & 15 & 1 \end{bmatrix}$$

Representation, 2:

- 2 hidden layers.
- $h_1 = \text{ReLU}(w_1 x + b_1)$ $\rightarrow$ Hidden layer 1
- $h_2 = \text{ReLU}(w_2 h_1 + b_2)$ $\rightarrow$ Hidden layer 2
- $o = (w_3 h_2 + b_3)$ $\rightarrow$ output layer

$$W_{0,1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix} \quad p_0 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad q_0 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

$$W_1 = W_{0,1} \cdot (p_0 q_0^T) = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 4 \\ 2 & 0 & -2 \end{bmatrix} \quad b_1 = [0 \ 1 \ -1].$$

$$W_{0,2} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & -2 \end{bmatrix} \quad p_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad q_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$W_2 = W_{0,2} \cdot (p_1 q_1^T) = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix},$$

$$b_2 = [1 \ -1 \ 2].$$

$$W_{0,3} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad q_2 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

$$W_3 = W_{0,3} \circ (P_2 q_2^T) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$b_3 = [2, 1, -1].$$

Passing the dataset through rep 2:

D = input dataset.

W_i = $i=1 \times 2 \Rightarrow$ Hidden layer i weights, $i=3$ = output layer weight

b_i = $i=1 \& 2 \Rightarrow$ Hidden layer i bias., $i=3$ = output layer bias.

O = output data

h_i = hidden layer i output.

$$\bullet h_1 = \text{ReLU}(W_1 D + b_1)$$

$$\bullet h_2 = \text{ReLU}(W_2 h_1 + b_2)$$

$$\bullet O = W_3 h_2 + b_3.$$

$$W_1 D^T + b_1 = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 4 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$
$$+ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & -8 & -10 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & -10 & -6 & 4 & 11 & -15 & 1 \\ -11 & -11 & -19 & -3 & 5 & 5 & 9 & 15 & -1 \end{bmatrix}.$$

$$h_1 = \text{ReLU}(W_1 D^T + b_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & 0 & 0 & 4 & 11 & 0 & 1 \\ 0 & 0 & 0 & 0 & 5 & 5 & 9 & 15 & 0 \end{bmatrix}.$$

$$W_2 h_1 + b_2 = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & 0 & 0 & 4 & 11 & 0 & 1 \\ 0 & 0 & 0 & 0 & 5 & 5 & 9 & 15 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 33 & 15 & 15 & 1 & -4 & 4 & 14 & -14 & 4 \\ -1 & -1 & -1 & -1 & -16 & -16 & -12 & 10 & -2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2) = \begin{bmatrix} 33 & 15 & 15 & 1 & 0 & 4 & 14 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$0 = W_3 h_2 + b_3 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 33 & 15 & 15 & 1 & 0 & 4 & 14 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$0 = \begin{bmatrix} 35 & 17 & 17 & 3 & 2 & 6 & 16 & 22 & 6 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -34 & -16 & -16 & -2 & -1 & -5 & -15 & -1 & -5 \end{bmatrix}$$

Representation 3:

- 2 hidden layers
- $h_1 = \text{ReLU}(W_1 x + b_1) \rightarrow \text{Hidden layer 1}$
- $h_2 = \text{ReLU}(W_2 h_1 + b_2) \rightarrow \text{Hidden layer 2}$
- $o = W_3 h_2 + b_3 \rightarrow \text{Output layer}$

$$W_{0,1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix} \quad p_0 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad q_0 = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} -2 & 0 & 0 \\ 4 & 0 & -4 \\ 4 & 0 & 2 \end{bmatrix} \quad b_1 = [1, -1, -1]$$

$$W_{0,2} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & -2 \end{bmatrix} \quad p_1 = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \quad q_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ -2 & -2 & -4 \end{bmatrix} \quad b_2 = [2, 2, 1]$$

$$W_{0,3} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad q_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$W_3 = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 2 & -2 \\ 1 & 0 & 0 \end{bmatrix} \quad b_3 = [2, -1, -1].$$

Passing the dataset through representation 3.

D = input dataset

weights

$w_i \Rightarrow i=1 \& 2 \Rightarrow$ Hidden layer i 's weights, $i=3 \Rightarrow$ output layer's

$b_i \Rightarrow i=1 \& 2 \Rightarrow$ Hidden layer i 's bias, $i=3 \Rightarrow$ output layer's bias.

O = output data

h_i = hidden layer i output.

$$\cdot h_1 = \text{ReLU}(w_1 D + b_1)$$

$$\cdot h_2 = \text{ReLU}(w_2 h_1 + b_2)$$

$$\cdot O = w_3 h_2 + b_3.$$

$$w_1 D^T + b_1 = \begin{bmatrix} -2 & 0 & 0 \\ 4 & 0 & -4 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 5 & 11 & 7 & 1 & -3 & -9 & -5 & 1 \\ -21 & -21 & -37 & -9 & 11 & 11 & 19 & 31 & -1 \\ 9 & -3 & -13 & -17 & -7 & 5 & 19 & 1 & -1 \end{bmatrix}$$

$$h_1 = \text{ReLU}(w_1 D^T + b_1) = \begin{bmatrix} 1 & 5 & 11 & 7 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 11 & 11 & 19 & 31 & 0 \\ 9 & 0 & 0 & 0 & 0 & 5 & 19 & 1 & 0 \end{bmatrix}$$

$$w_2 h_1 + b_2 = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ -2 & -2 & -4 \end{bmatrix} \begin{bmatrix} 1 & 5 & 11 & 7 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 11 & 11 & 19 & 3 & 0 \\ 9 & 0 & 0 & 0 & 0 & 5 & 19 & 1 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & 2 & 2 & 2 & 24 & 34 & 78 & 66 & 2 \\ 13 & 12 & 24 & 16 & 4 & 7 & 21 & 3 & 4 \\ -37 & -9 & -21 & -13 & -23 & -41 & -113 & -65 & -1 \end{bmatrix}$$

$$h_2 = \text{ReLU}(w_2 h_1 + b_2) = \begin{bmatrix} 20 & 2 & 2 & 2 & 24 & 34 & 78 & 66 & 2 \\ 13 & 12 & 24 & 16 & 4 & 7 & 21 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$0 = w_3 h_2 + b_3 = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 2 & -2 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 20 & 2 & 2 & 2 & 24 & 34 & 78 & 66 & 2 \\ 13 & 12 & 24 & 16 & 4 & 7 & 21 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & -20 & -44 & -28 & 18 & 22 & 38 & 62 & -4 \\ -15 & 19 & 43 & 27 & -41 & -95 & -115 & -127 & 3 \\ 19 & 1 & 1 & 1 & 1 & 23 & 53 & 77 & 65 \end{bmatrix}$$